

torch.all#

<https://pytorch.org/docs/stable/generated/torch.all.html>

Description:

Tests if all elements in input evaluate to True. This function matches the behaviour of NumPy in returning output of dtype bool for all supported dtypes except uint8. For uint8 the dtype of output is uint8 itself. input (Tensor) – the input tensor. out (Tensor, optional) – the output tensor.

Function Signature

```
torch.all(input: Tensor, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

```
torch.all(input, dim, keepdim=False, *, out=None) → Tensor
```

Parameters

Parameters

Keyword

out (Tensor, optional) – the output tensor.

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out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.rand(1, 2).bool()
>>> a
tensor([[False, True]], dtype=torch.bool)
>>> torch.all(a)
tensor(False, dtype=torch.bool)
>>> a = torch.arange(0, 3)
>>> a
tensor([0, 1, 2])
>>> torch.all(a)
tensor(False)
```

```
>>> a = torch.rand(4, 2).bool()
>>> a
tensor([[True, True],
        [True, False],
        [True, True],
        [True, True]], dtype=torch.bool)
>>> torch.all(a, dim=1)
tensor([ True, False,  True,  True], dtype=torch.bool)
>>> torch.all(a, dim=0)
tensor([ True, False], dtype=torch.bool)
```

Notes & Warnings

NOTE

Note This function matches the behaviour of NumPy in returning output of dtype bool for all supported dtypes except uint8. For uint8 the dtype of output is uint8 itself.

Detailed Documentation

Documentation

Tests if all elements in input evaluate to True.

This function matches the behaviour of NumPy in returning output of dtype bool for all supported dtypes except uint8. For uint8 the dtype of output is uint8 itself.

input (Tensor) – the input tensor.

out (Tensor, optional) – the output tensor.

For each row of input in the given dimension dim, returns True if all elements in the row evaluate to True and False otherwise.

If keepdim is True, the output tensor is of the same size as input except in the dimension(s) dim where it is of size 1. Otherwise, dim is squeezed (see torch.squeeze()), resulting in the output tensor having 1 (or len(dim)) fewer dimension(s).

input (Tensor) – the input tensor.

dim (int or tuple of ints, optional) – the dimension or dimensions to reduce. If None, all dimensions are reduced.

keepdim (bool, optional) – whether the output tensor has dim retained or not. Default: False.

out (Tensor, optional) – the output tensor.
